AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 13 · TEACHER GUIDE

State machines for bounded agents

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which control authorizes each move and what evidence remains unverified.

Use the original graph and copied revision to distinguish spare budget from a permitted retry, and valid prefixes from completed tasks.

Keep one habit: Model first-invalid-edge reasoning → inspect separate guards → compare explicit graph versions → assess fresh narrower paths → preserve rejected traces.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **state** machines for bounded agents

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of logs traces and privacy

Reference example, not the only acceptable wording: Teacher explanation: Logs support diagnosis, but excessive recording can expose private content. Trace IDs link stages without storing complete sensitive **input**s.

For this policy log trace ID, stage, status, duration and error category; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of evaluation pipelines

Reference example, not the only acceptable wording: Teacher explanation: Evaluation pipelines preserve case versions, procedure versions and raw outcomes before producing metrics. A rerun must identify what changed.

Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of testing pure components

Reference example, not the only acceptable wording: Teacher explanation: Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

Pictured end goal

Explain **state** machines for bounded agents using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Case exposure check: this source example appears in ai-2/30, ai-6/13. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Logs traces and privacy

Evaluation pipelines

Testing pure components

Use the three-session teaching cycle above, with extra time as needed. A bounded agent uses explicit **state**s and finite action limits. A **state** machine makes invalid **transition**s visible. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **state** machines for bounded agents

Apply the **state**d procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach transition order, evidence guards and action budgets as separate controls

> Keep original, revised, parcel and selected independent graphs distinct.

Definitions and complete original policy

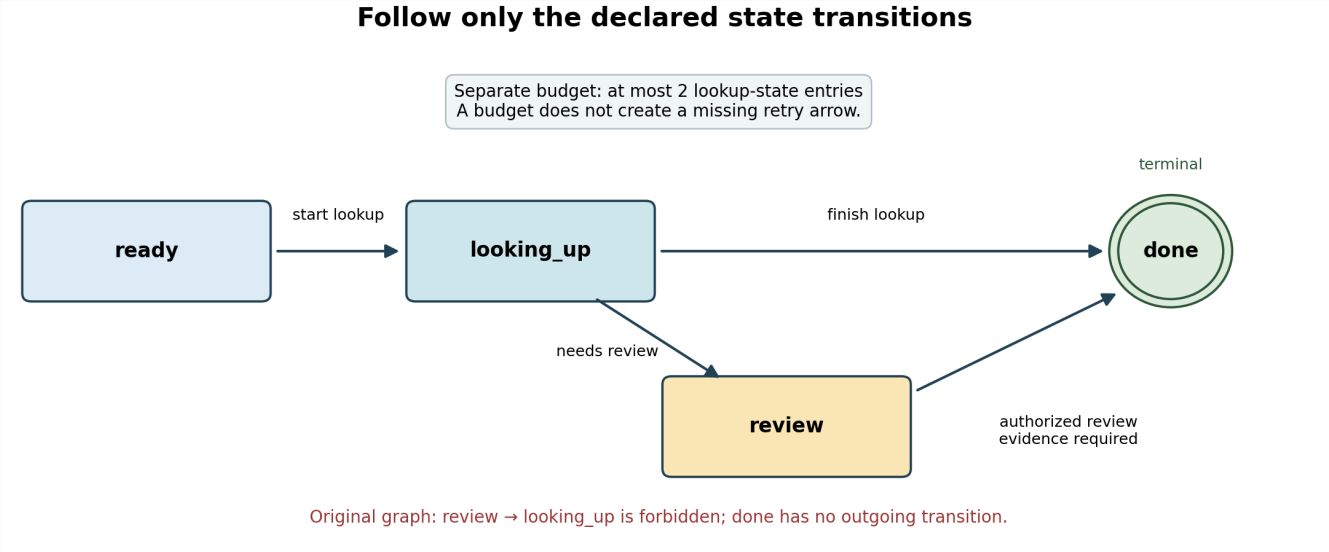
A bounded agent uses explicit **state**s and finite action limits. A **state** machine makes invalid **transition**s visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Main original state policy	Exact rule
Start	ready, with lookup-entry counter 0
ready	May move only to looking_up
looking_up	May move to review or done
review	May move to done only, after authorized review evidence
done	Terminal: no outgoing transition
Lookup budget	At most 2 accepted entries into looking_up; does not add retry arrows

Main original state policy	Exact rule
Valid prefix	Every transition so far is allowed; may still be incomplete in review
Evidence scope	Local boolean evidence flag is a supplied trusted fixture, not real evidence verification
Original graph consequence	No retry cycle exists, so a valid original path enters looking_up at most once

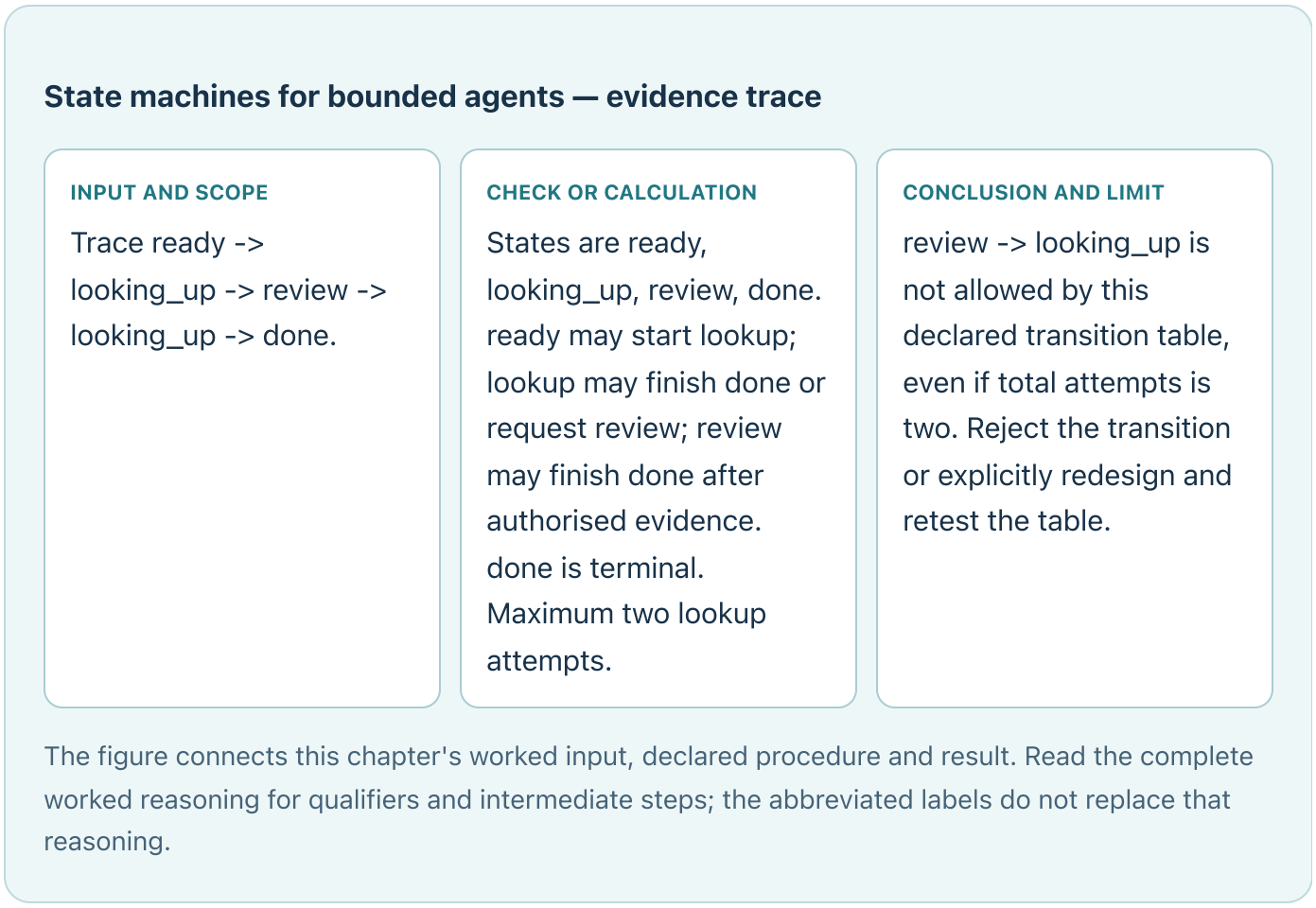
Directed graph, invalid-prefix trace and checked reasoning



Proposed adjacent step	Arrow check	Accepted lookup entries	Result
ready → looking_up	Allowed	1	Continue
looking_up → review	Allowed	1	Continue; review is not done
review → looking_up	Forbidden in original graph	1	Reject before second entry
Remaining proposed → done	Not processed after rejection	Counter remains 1	Do not report the whole trace valid
Two lookup markers in proposed text	Budget alone would not exceed 2	Not two accepted executions	Missing arrow still invalidates proposal

Main forbidden edge is review→looking_up, even with unused budget. The accepted prefix contains one lookup entry; the proposed text contains two lookup markers, not two authorized executions. Original graph has no retry loop and admits at most one entry. Teach review→done guard separately: the code accepts a supplied trusted evidence flag, not verified real evidence.

The selected current independent task supplies a narrower graph ready→looking_up→review→done; assess its explicit adjacency table without borrowing the main direct looking_up→done edge. Older terminal-restart practice is reference material, not the current selected case.



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Qualification: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

Exit: accept a concrete variation with its predicted outcome and a **state**d condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Trace ready -> looking_up -> review -> looking_up -> done.

review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Compare the intermediate **evidence** with the final **state**ment. Explain why each number or decision follows from the **state**d **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Guard, budget and terminal-case probes

Complete transition/budget probe	Expected interpretation
Original ready→looking_up→review	Valid prefix, incomplete task
Then review→done without authorized review evidence	Declared arrow exists but evidence guard rejects
Same path with supplied authorized-evidence flag	Graph/guard audit can complete; flag is not independently verified evidence
Original review→looking_up with spare budget	Forbidden arrow; spare budget does not authorize it
Explicit revised graph adds review→looking_up	Retry edge can be used, subject to retained budget
Revised graph attempts a third lookup entry	Budget rejects even though retry edge exists
done→looking_up	Terminal-state policy rejects
Compact original helper accepts an arrow	It checks only adjacency; complete budget/evidence checks are in the new audit
A graph audit passes	Does not prove real tool execution, fetched facts or external correctness

Separate parcel graph and scope

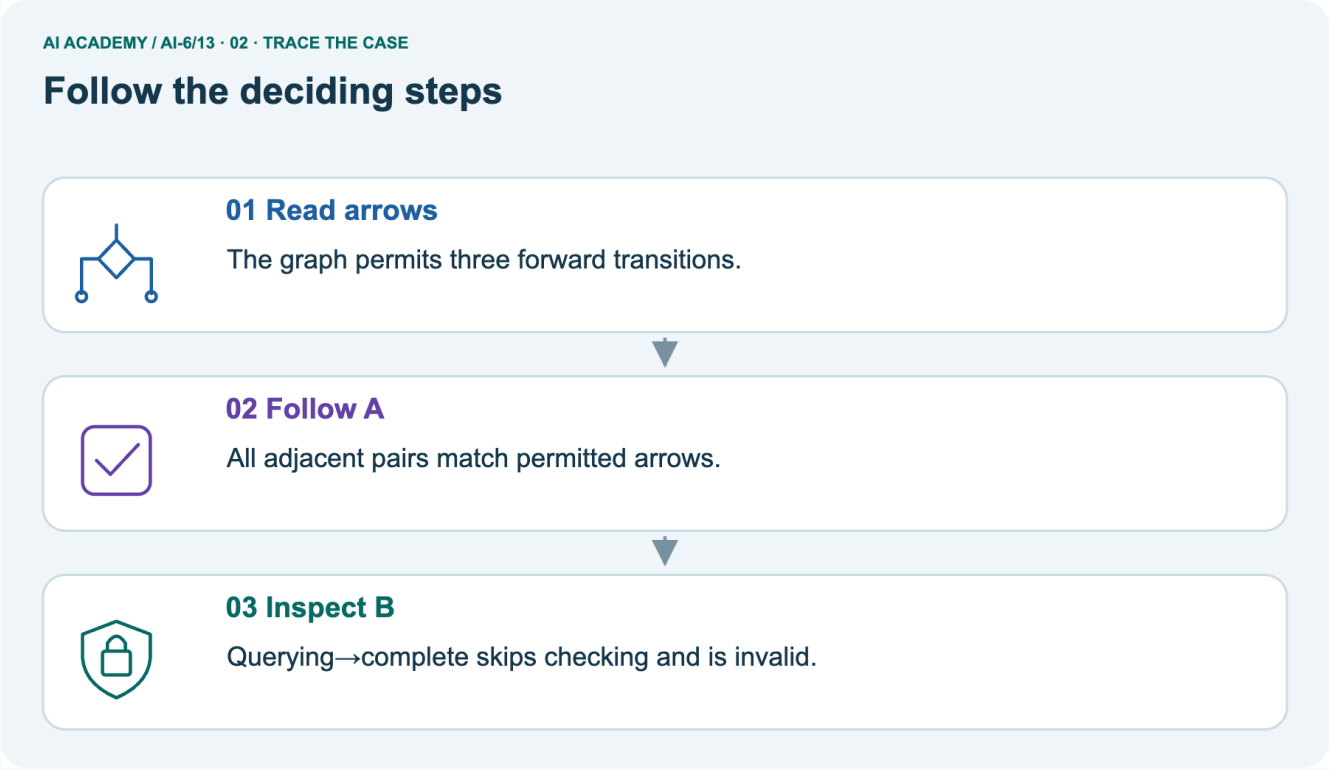
A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal.

A follows all four **state**s. B follows idle→querying→complete. No other **transition**s are declared.

The main ready/looking_up/review/done graph and the picture's parcel graph are separate. Parcel permits idle→querying→checking→complete, with complete terminal. Path A uses all four states and follows every declared arrow; B goes querying→complete and skips checking, so it is invalid. Adding checking→querying permits that one retry edge but still does not permit querying→complete.

A real retrying agent also needs an action budget; the picture only checks its stated finite paths, not parcel-data truth or a running autonomous loop.

Parcel trace and one-arrow revision key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read arrows	The graph permits three forward transitions.
2. Follow A	All adjacent pairs match permitted arrows.
3. Inspect B	Querying→complete skips checking and is invalid.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **state** machine bounds action order. A plausible final **state** does not establish that the path was valid.

This graph checks sequence, not the correctness of the fetched parcel data.

Change: Add checking→querying as a permitted retry.

Prediction: That retry becomes valid; querying→complete stays forbidden.

Reason: Adding one arrow does not authorise unrelated shortcuts.

▼ Reference: original adjacency-only code

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and **state** the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def transition(state, target):
    allowed = {'ready': {'looking_up'}, 'looking_up': {'review', 'done'}, 'review':
{'done'}, 'done': set()}
    if target not in allowed.get(state, set()):
        raise ValueError('invalid transition')
    return target

print(transition("ready", "looking_up"))
```

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the **state**d contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
State	A declared stage of a process
Transition	An allowed move between states
Terminal state	A state with no further operations under the contract

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

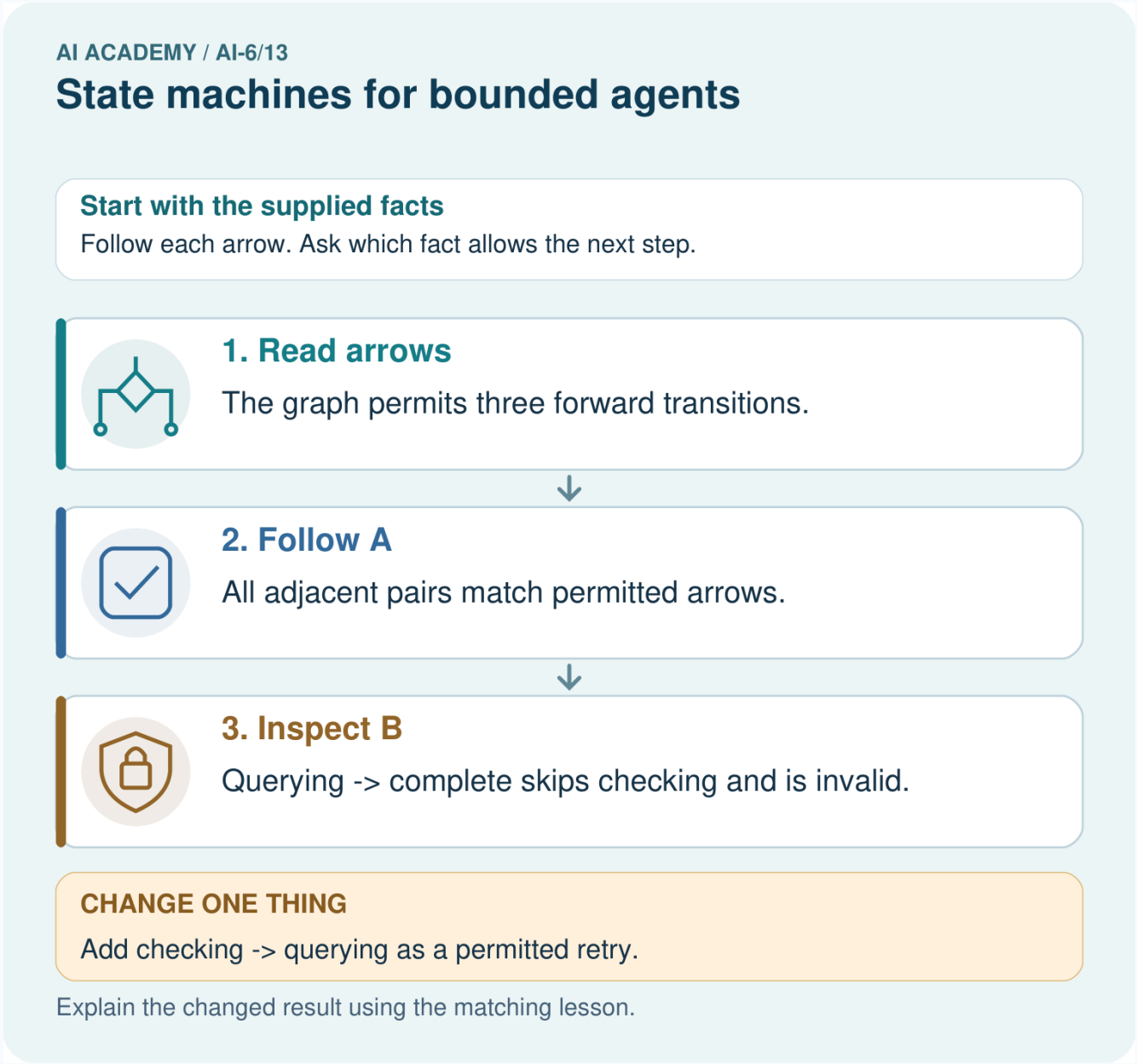
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: State machines for bounded agents

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Trace ready -> looking_up -> review -> looking_up -> done. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **State** machines for bounded agents. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Trace A and B through the **state** diagram. Name B's forbidden **transition** and explain whether adding checking→querying permits that shortcut.

The graph permits three forward **transition**s. All adjacent pairs match permitted arrows. Querying→complete skips checking and is invalid. Changed case: That retry becomes valid; querying→complete stays forbidden.

Adding one arrow does not authorise unrelated shortcuts.

Changed-case reasoning: That retry becomes valid; querying→complete stays forbidden. Adding one arrow does not authorise unrelated shortcuts.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main original retry from review invalid despite cap 2; separate graph revision and budget guard needed. Review completion guard/evidence and valid-prefix versus complete distinction required. Parcel B skips checking; adding reverse retry edge does not legalize the shortcut.

Selected narrower graph: ready→review→done invalid at ready→review; ready→looking_up→review→done is structurally valid under its stated arrows, with done terminal. It supplies no real evidence-verification outcome. Use selected key, not earlier terminal-restart wording.

Diagnose the current misconceptions

The worked result establishes every future case: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Only these **transition**s are allowed: ready→looking_up→review→done. Done is terminal. Evaluate ready→review→done and ready→looking_up→review→done.

Teacher explanation: The first trace fails at ready→review because that arrow is absent from this case's allowed **transition**s. Do not continue an invalid trace as though it passed. The second trace ready→looking_up→review→done follows every declared arrow and ends at the terminal **state**. No **transition** may leave done.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A bounded agent uses explicit **state**s and finite action limits. A **state** machine makes invalid **transition**s visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

W2 Trace the worked case

Trace ready -> looking_up -> review -> looking_up -> done. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

W3 Audit the limit

A report omits this qualification: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

The new case should test this particular **mechanism** rather than merely re**state** the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Qualification: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

Exit: accept a concrete variation with its predicted outcome and a **state**d condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid **transition**.

Teacher explanation: done -> looking_up violates terminal-**state** rule; stop. Earlier **transition**s are permitted.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: done -> looking_up violates terminal-**state** rule; stop. Earlier **transition**s are permitted. Require intermediate reasoning and the **state**d scope. Fresh retry: ready -> looking_up -> done -> review.

Expected reasoning: done -> review invalid. A **terminal state** must not silently resume. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **state** machines for bounded agents **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **state** machines for bounded agents.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Planning and action selection. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/14, ai-6/16, ai-6/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **state** machines for bounded agents

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Trace ready -> looking_up -> review -> looking_up -> done. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Check every learner before discussion: Use the worked case for **State** machines for bounded agents. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add checking→querying as a permitted retry. Require a **prediction** and a reason before comparing. Expected result: That retry becomes valid; querying→complete stays forbidden.

Reason: Adding one arrow does not authorise unrelated shortcuts.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

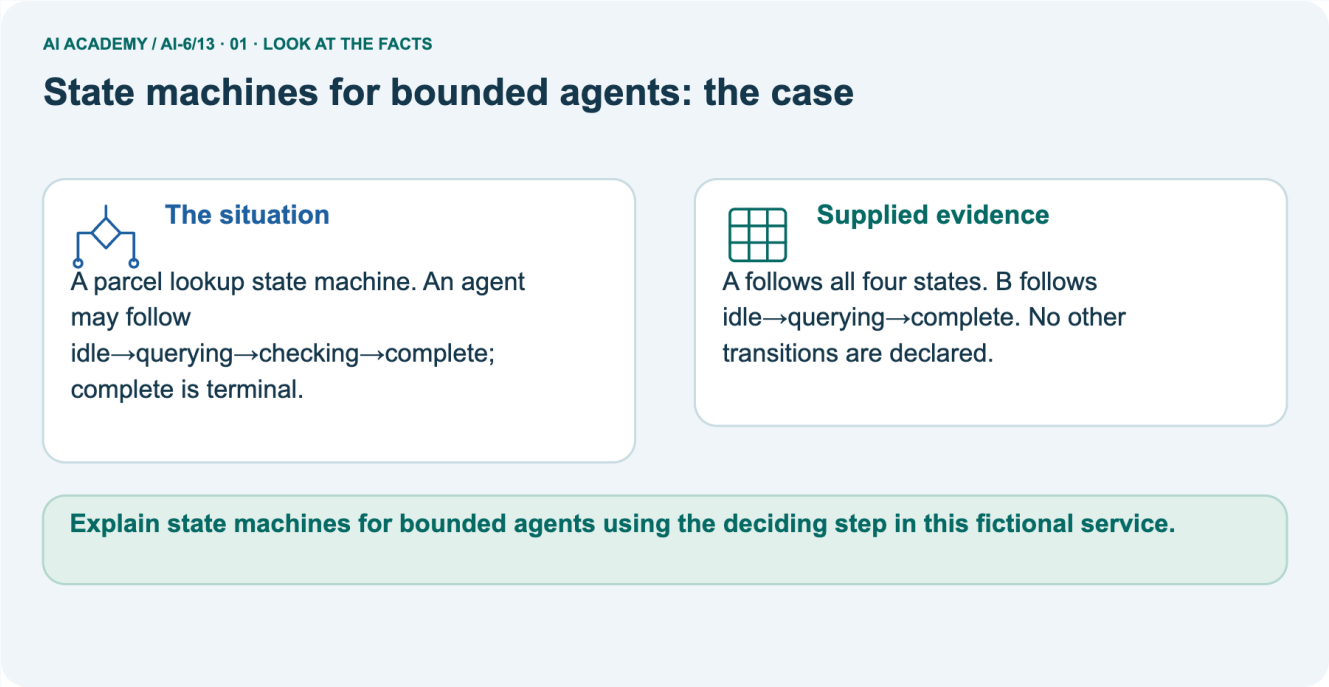
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. State machines for bounded agents: the case



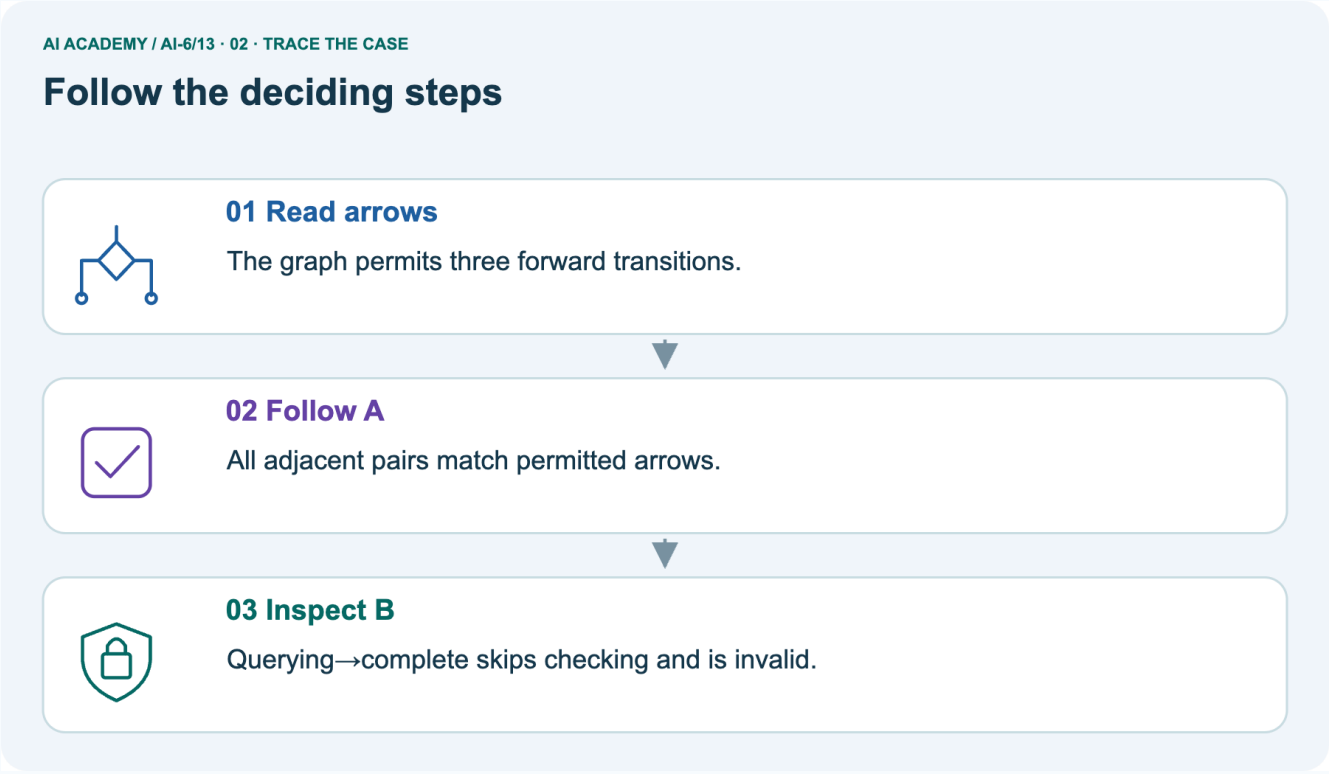
Follow the labelled picture with the matching explanation. Open the vector diagram

A parcel lookup **state** machine. An agent may follow idle→querying→checking→complete; complete is terminal. Supplied facts: A follows all four **state**s. B follows idle→querying→complete. No other **transition**s are declared.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



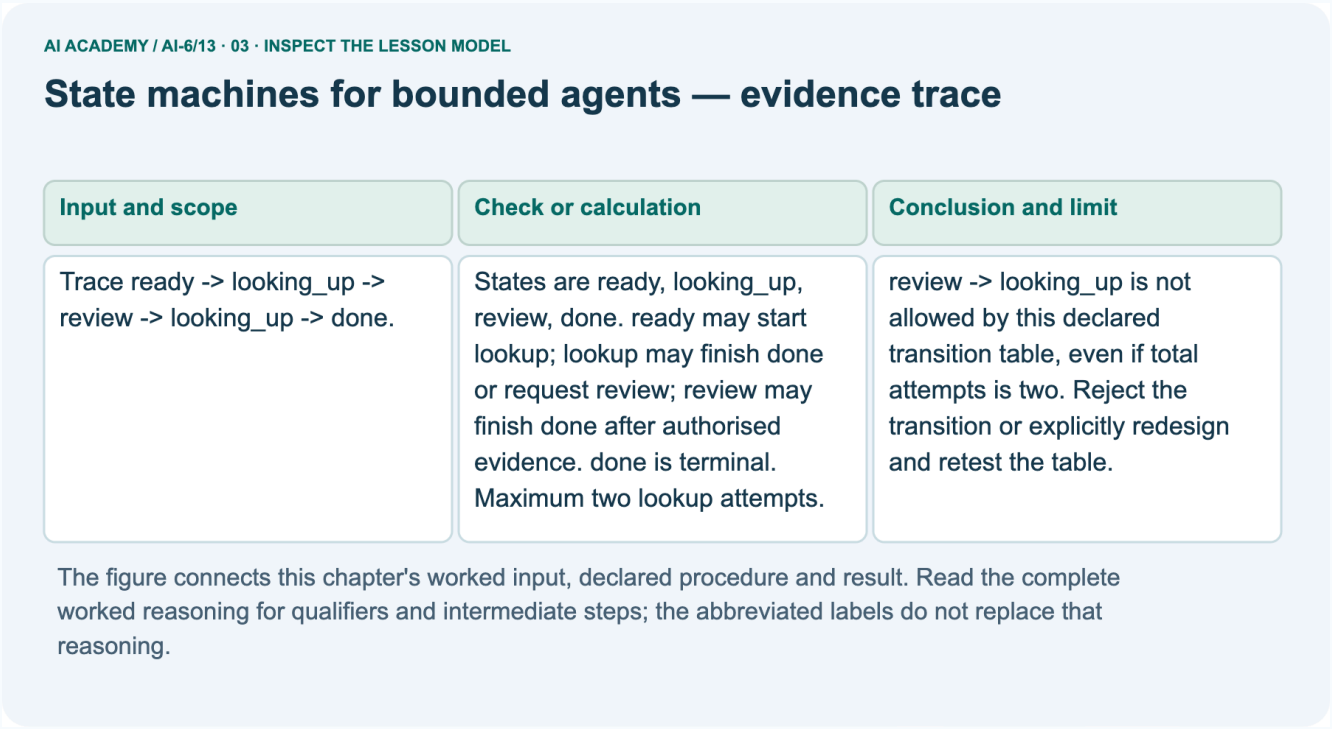
Follow the labelled picture with the matching explanation. Open the vector diagram

Read arrows: The graph permits three forward **transition**s. → Follow A: All adjacent pairs match permitted arrows. → Inspect B: Querying→complete skips checking and is invalid.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. State machines for bounded agents — evidence trace



AI ACADEMY / AI-6/13 · 03 · INSPECT THE LESSON MODEL

State machines for bounded agents — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Trace ready -> looking_up -> review -> looking_up -> done.	States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.	review -> looking_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

State machines for bounded agents — **evidence** trace.

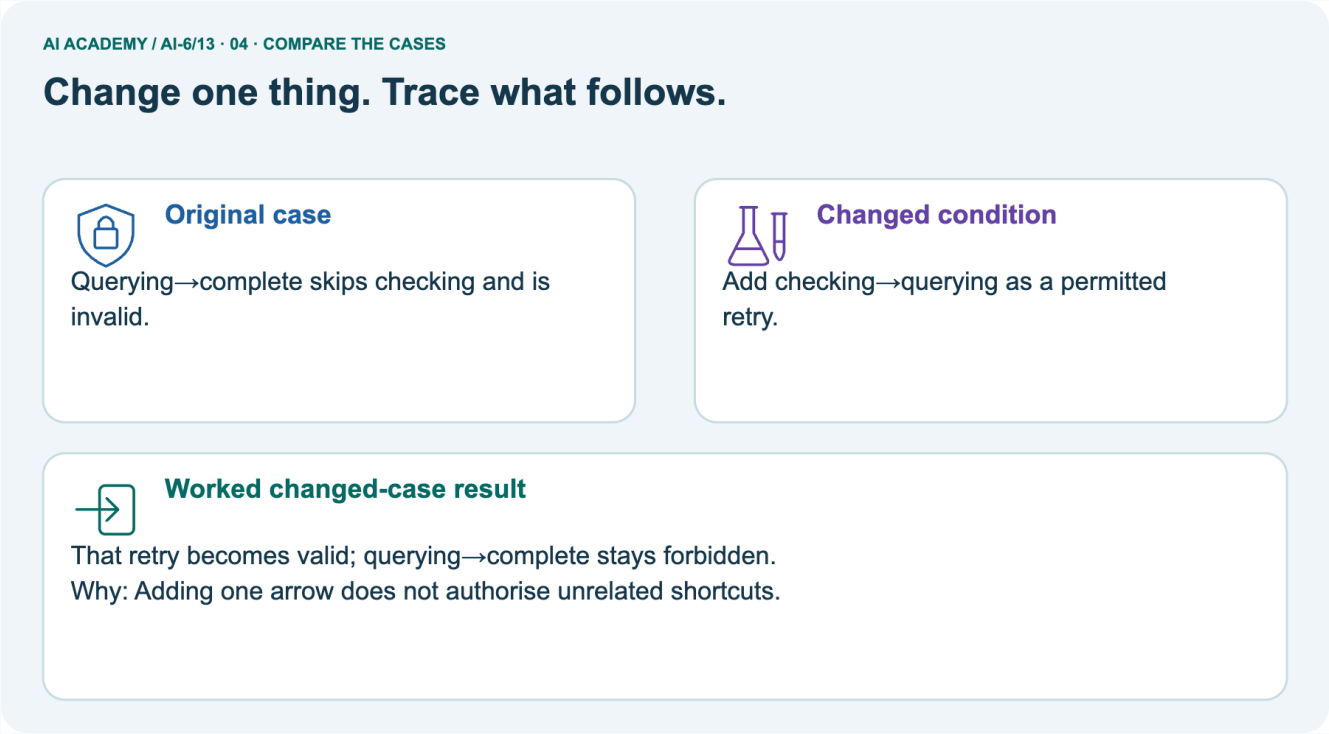
Trace ready -> looking_up -> review -> looking_up -> done. | **State**s are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal.

Maximum two lookup attempts. | review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Querying→complete skips checking and is invalid. Changed condition: Add checking→querying as a permitted retry. Expected result: That retry becomes valid; querying→complete stays forbidden.

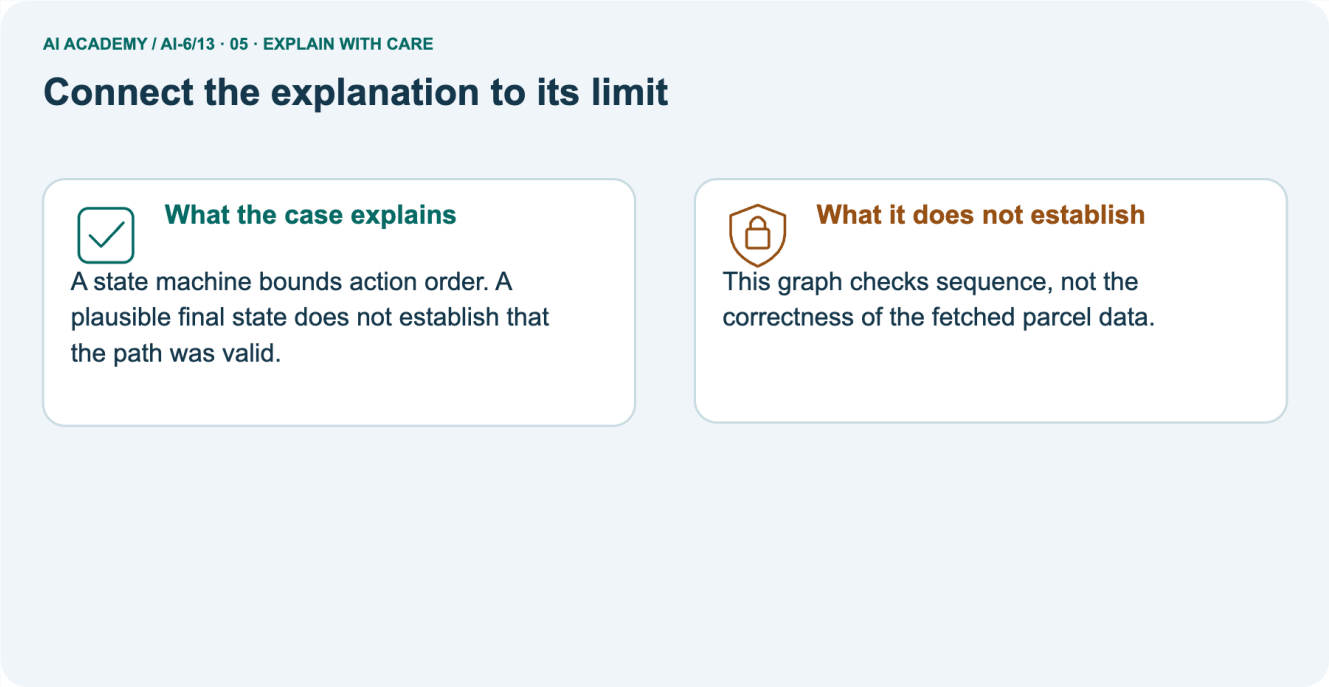
Reason: Adding one arrow does not authorise unrelated shortcuts.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Adding one arrow does not authorise unrelated shortcuts.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

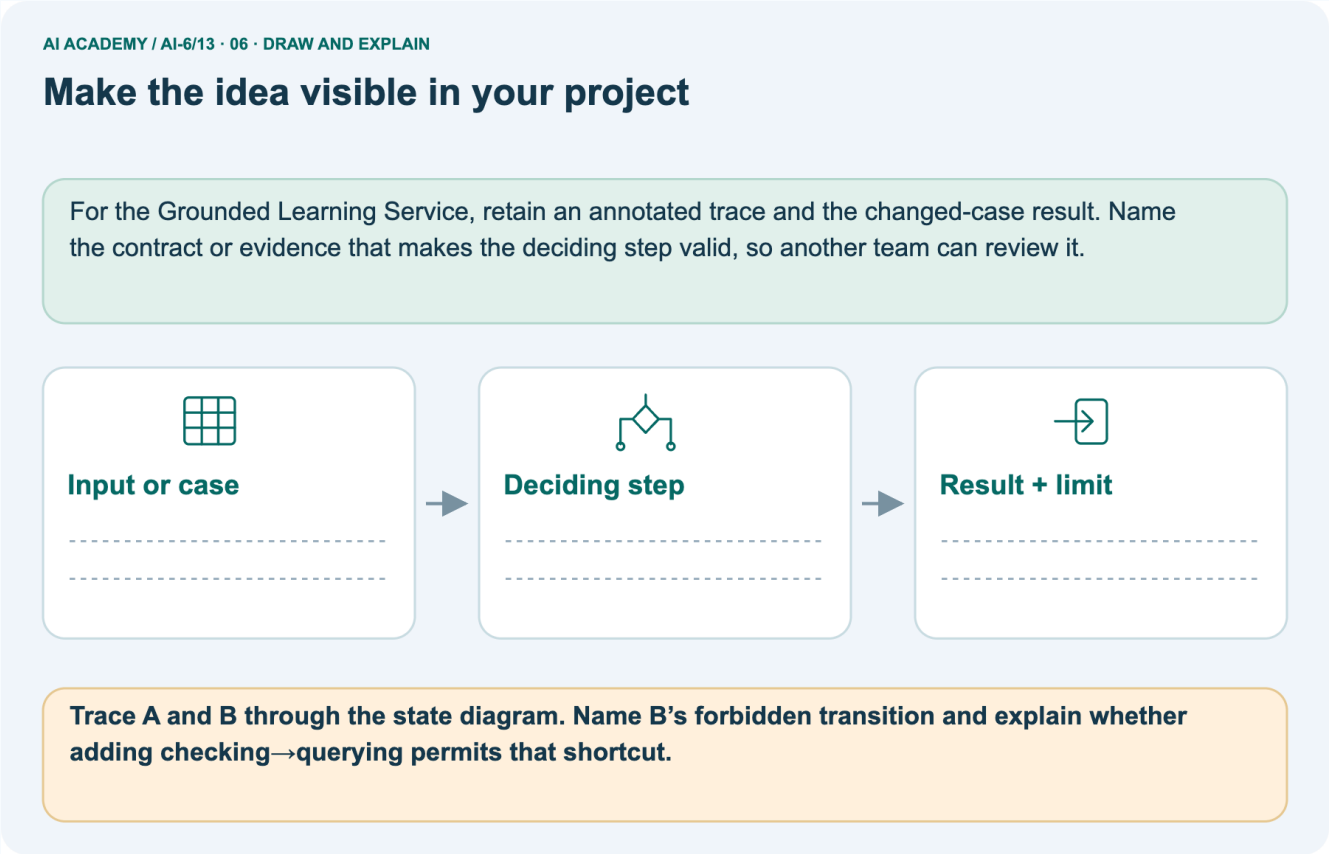
A **state** machine bounds action order. A plausible final **state** does not establish that the path was valid. Limit: This graph checks sequence, not the correctness of the fetched parcel data.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the **state**d scope: This graph checks sequence, not the correctness of the fetched parcel data.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace A and B through the **state** diagram. Name B's forbidden **transition** and explain whether adding checking→querying permits that shortcut.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The graph permits three forward **transition**s. All adjacent pairs match permitted arrows. Querying→complete skips checking and is invalid.

Changed case: That retry becomes valid; querying→complete stays forbidden. Adding one arrow does not authorise unrelated shortcuts.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. State machines for bounded agents: the case

02. Follow the deciding steps

03. State machines for bounded agents — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.